Patient Voice Assistant

A live voice booking assistant for a mock clinic, built on Boson AI Higgs APIs. It books care, and when it's an emergency, it puts the patient first automatically.

Higgs APIs used

API	How it's used
Higgs Realtime	Full-duplex voice conversation with Ava: server VAD, barge-in, live captions. The app controls when Ava replies, so every answer uses the latest clinic data.
Higgs STT 3.1	Transcribes the patient inside the Realtime session; the transcript drives the safety net, booking logic and detail capture.
Higgs TTS 3	Voices demo phrases into the live session as patient audio, and is Ava's voice in offline mode.

Normal booking flow

- Patient describes the problem, then a **symptom** → **specialty** → **doctor** match (General Practice, Cardiology, Dermatology, Orthopedics, Pediatrics).
- Ava offers the **three nearest open times** from the current date and time; understands "Monday", "October 28th", "tomorrow afternoon", "after 3 PM", "the first one".
- Questions like "Do you have 3 PM Wednesday?" get a confirmation first; a picked time is **held for 5 minutes**.
- Collects **name** → **address** → **phone** one at a time (address taken as said; no city or postal-code follow-ups), shown in a highlighted details box.
- **Final check**: Ava reads back all details; "yes" opens the **payment screen**, and the chat pauses until **Done** or **Release hold**; Ava then confirms, and a receipt SMS arrives.
- Changing the day or time while a slot is held moves the hold and keeps the details already given.
- **Rescheduling** a confirmed booking: found from this chat, the booking ID ("P V A 9 6 5 W") or the phone number; Ava offers new times, confirms, then moves it. Same booking ID, payment carries over, old slot freed, SMS with the new time; "never mind" keeps the original.

Emergency flow

- **Safety net**: red-flag keywords checked on the live (even partial) transcript, e.g. chest pain, can't breathe, stroke signs, heavy bleeding, choking, poisoning, head injury, dislocations, baby with fever, suicidal thoughts. Negations like "not urgent" or "urgent care" are ignored.
- Also triggered by Ava herself or by a manual **Force alert**; the banner shows which source raised it.
- **Triage**:
  - **Monday–Saturday**: the nearest slot today, **pre-empting** an existing booking if needed (that patient is moved to the next free slot and texted); if the specialist is off, a working doctor covers.
  - **Sunday or after hours**: "call 911 / go to the nearest ER now", plus a priority follow-up at the next working slot.
- Routing: child → Pediatrics, chest/heart → Cardiology, bones/dislocations → Orthopedics, otherwise GP; uses the whole conversation, not just the trigger sentence.
- No hold or payment; the screen turns red and the emergency SMS is sent.

Clinic calendar and data

- **Real calendar** from the current date and time, 90 days ahead: Monday–Saturday, 9–12 and 2–4, 30-minute slots, closed Sundays. Each doctor has fixed pseudo-random days off and blocked times.
- **Availability calendar**: pick a doctor, browse months, click a day to see open/booked/unavailable times, click an open time to book.

- **Bookings database** (`data/bookings.json`): every booking with patient details, problem, specialty, doctor, dated slot, payment, emergency triage, pre-emption, SMS and full status history; survives resets and restarts.
- **Conversation history** (`data/conversations/`): every chat saved under an ID; a page refresh restores it, a History picker reopens past chats, and Ava resumes where she left off.

Interface

- Character with four states (idle, listening, speaking, alert); Doctor / Time / Booking ID / Payment cards; clinic dashboard with live schedule and activity log.
- Live captions with keyword highlighting: doctors and times blue, payment amber, emergency red.
- Virtual phone showing SMS receipts and emergency alerts; live clock; session timer with cost estimate.
- **Demo mode**: canned phrases or typed input, live (voiced into the session) or offline (scripted replies); Reset demo; backup video.

Reliability work

- App-controlled turn-taking (`create_response: false`) fixes Ava answering before the transcript arrived.
- **Higgs pipeline strip** under the captions shows each turn's STT time, agent decision and Realtime first-audio time (about 0.9 s end of speech to Ava speaking).
- Guards correct Ava if she asks for city/street details or names a time that differs from the screen.
- Speech cleanup: digit words to numbers, phone numbers joined across pauses, filler words removed from names and addresses.
- 50 automated tests (calendar, emergency triage, rescheduling, bookings DB, conversations, keyword parsing); zero npm dependencies, Node 18+.

Code: github.com/Shubh8434/hackathon-higgs (patient-voice-assistant/, tag submission) · Run: `npm start` → http://localhost:3100 · See also Higgs_API_Feedback_Report.pdf